

CPU SCHEDULING SIMULATOR

TEST CASE DOCUMENTATION

1. PROJECT OVERVIEW

This project is a web-based simulator designed to compare two fundamental CPU scheduling algorithms:

- **Preemptive Priority Scheduling:** A policy-driven approach focusing on task importance.
- **Shortest Remaining Time First (SRTF):** An efficiency-driven approach focusing on minimizing completion time.

The goal is to analyze how different workloads behave under these two distinct scheduling strategies in terms of **Waiting Time**, **Turnaround Time**, and **Response Time**.

2. SCHEDULING SPECIFICATIONS

- **Priority Rule:** Lower numeric value = Higher Priority (e.g., 1 is higher than 5).
- **Tie-breaking:** FCFS (First Come First Served) based on Arrival Time.
- **Preemption:** Both algorithms are fully preemptive.

3. DETAILED TEST SCENARIOS

SCENARIO A: BASIC MIXED WORKLOAD

Objective: Evaluate baseline behavior with a balanced set of processes.

PID	Arrival Time	Burst Time	Priority
1	0	8	3
2	1	4	1
3	2	9	4
4	3	5	2
5	5	2	5

SCENARIO B: THE PRIORITY CONFLICT

Objective: Test how a high-priority long process interacts with low-priority short processes.

PID	Arrival Time	Burst Time	Priority
1	0	12	1
2	2	2	4
3	3	3	3
4	5	1	5

SCENARIO C: STARVATION SENSITIVITY

Objective: Demonstrate "Starvation" where a low-priority process (PID 4) waits significantly longer because higher priority tasks keep arriving.

PID	Arrival Time	Burst Time	Priority
1	0	3	1
2	1	3	1
3	2	3	1
4	0	25	5
5	4	3	1

SCENARIO D: SYSTEM VALIDATION & ERROR HANDLING

Objective: Verify the simulator's ability to detect and reject invalid inputs.

- **PID 1:** Priority = 2.7 (Invalid: Must be integer).
- **PID 2:** Burst Time = 0 (Invalid: Must be ≥ 1).
- **PID 3:** Arrival Time = -1 (Invalid: Must be ≥ 0).
- **PID 4:** Normal process to verify partial execution block.

4. INPUT VALIDATION RULES

The system enforces the following constraints:

- **Arrival Time:** Must be a non-negative integer.
- **Burst Time:** Must be a positive integer (≥ 1).
- **Priority:** Must be a positive integer (≥ 1).
- **Process ID:** Must be unique and a positive integer.

5. CONCLUSION & ANALYSIS

By running these scenarios, users can observe the "Policy vs. Efficiency" trade-off. While **SRTF** typically minimizes average waiting time, **Priority Scheduling** ensures that critical (High Priority) tasks are never delayed by longer, less important jobs.